

CYBERSECURITY INCIDENT RESPONSE

Jira-Based Lab Documentation

Submitted By:

Hadiya Zuberi

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Project 1: Emotet infection with Qakbot(PCAP File)

Overview

Objective

The objective of this lab is to analyze the pcap file “**Emotet infection with Qakbot**” Emotet with Qakbot is a two-stage malware infection - very sophisticated.

- **Emotet** = Initial dropper (downloads Qakbot)
- **Qakbot** = Banking trojan (steals credentials, spreads)

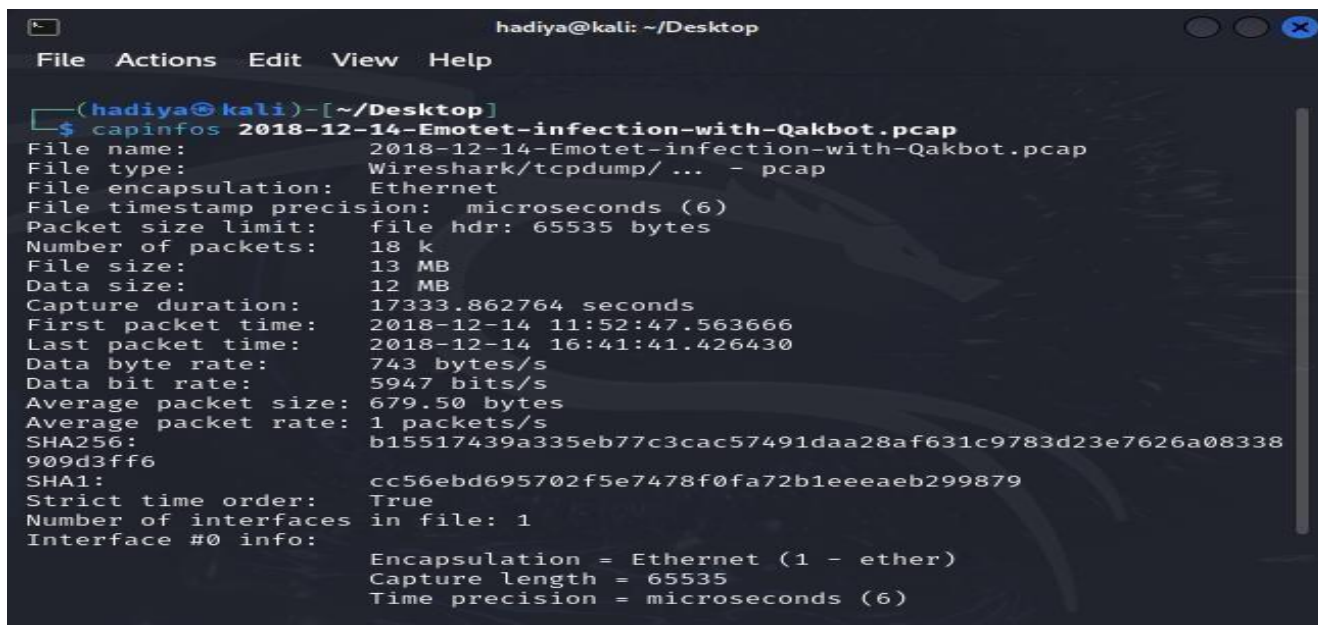
Tools Used

- Wireshark: for opening and analyzing the PCAP file
- Emotet infection with Qakbot pcap file : pre-captured network traffic file containing the attack
- Jira : for creating and managing the incident ticket

Lab Environment

The PCAP file was downloaded from netresec.com and opened locally in Wireshark. No virtual machines were required for this task. The target IP observed in the capture was 10.12.14.101, receiving traffic on port 443.

Initial Scan



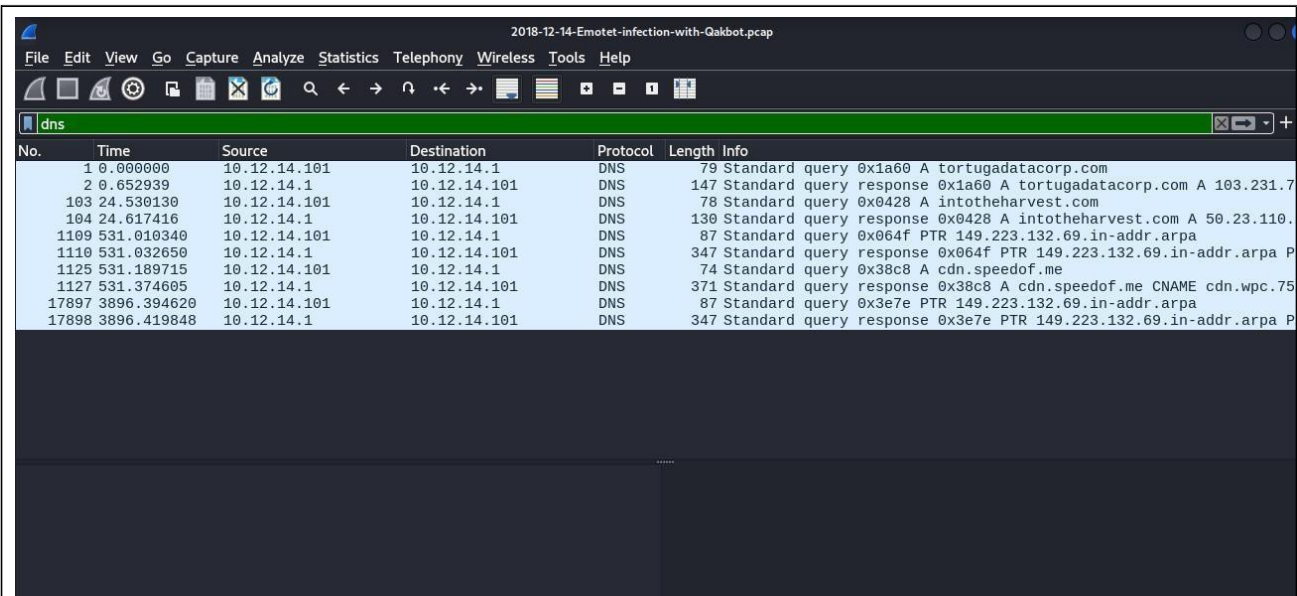
```
hadiya@kali: ~/Desktop
File Actions Edit View Help

(hadiya@kali)-[~/Desktop]
$ capinfos 2018-12-14-Emotet-infection-with-Qakbot.pcap
File name: 2018-12-14-Emotet-infection-with-Qakbot.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 18 k
File size: 13 MB
Data size: 12 MB
Capture duration: 17333.862764 seconds
First packet time: 2018-12-14 11:52:47.563666
Last packet time: 2018-12-14 16:41:41.426430
Data byte rate: 743 bytes/s
Data bit rate: 5947 bits/s
Average packet size: 679.50 bytes
Average packet rate: 1 packets/s
SHA256: b15517439a335eb77c3cac57491daa28af631c9783d23e7626a08338
909d3ff6
SHA1: cc56ebd695702f5e7478f0fa72b1eeeaeb299879
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
Encapsulation = Ethernet (1 - ether)
Capture length = 65535
Time precision = microseconds (6)
```

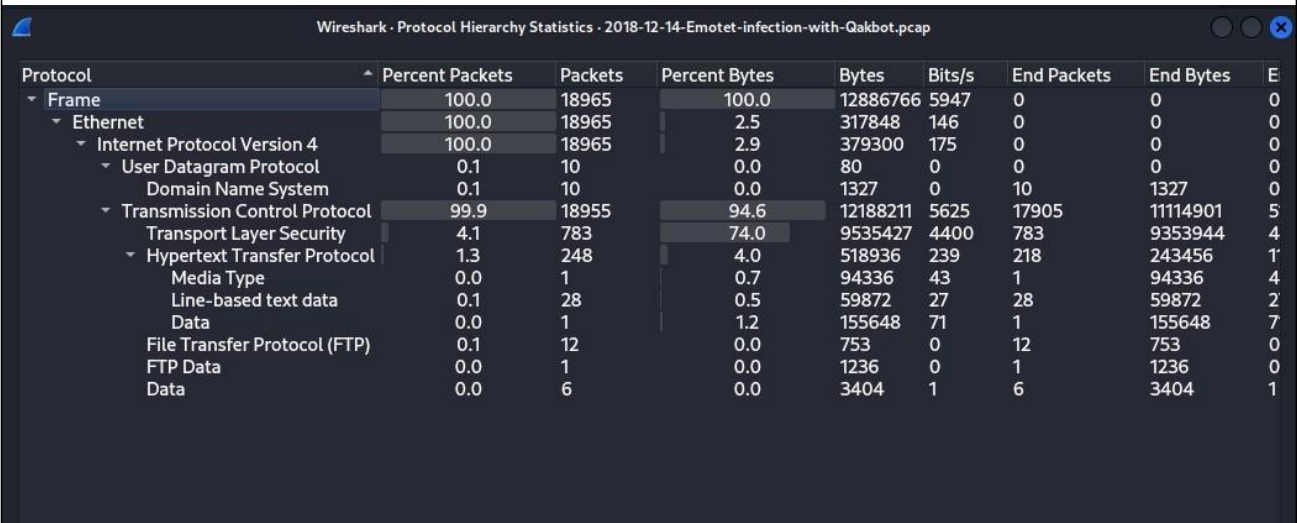
Picture 1.1: Initial Scan on Kali

Attack Simulation

Wireshark Traffic Capture



Picture 1.2 Packet analysis using filters for DNS



Picture 1.3: Protocol Hierarchy

Wireshark - Conversations - 2018-12-14-Emotet-infection-with-Qakbot.pcap

Conversation Settings

Name resolution

Absolute start time

Limit to display filter

Copy

Follow Stream...

Graph...

Protocol

Bluetooth

BPv7

DCCP

✓ Ethernet

FC

FDDI

IEEE 802.11

IEEE 802.15.4

✓ IPv4

✓ IPv6

Filter list for specific type

Ethernet · 1	IPv4 · 23	IPv6	TCP · 122	UDP · 5						
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
10.12.14.101	49217	72.21.81.189	443	6,588	5 MB	9	3,307	199 kB	3,281	4 MB
10.12.14.101	49226	72.21.81.189	443	6,585	5 MB	18	3,307	199 kB	3,278	4 MB
10.12.14.101	49222	73.233.236.181	443	1,321	900 kB	14	665	41 kB	656	859 kB
10.12.14.101	49263	81.136.148.196	50000	715	588 kB	62	283	19 kB	432	569 kB
10.12.14.101	49206	190.146.201.54	80	630	493 kB	3	269	18 kB	361	474 kB
10.12.14.101	49218	73.233.236.181	443	325	218 kB	10	165	11 kB	160	207 kB
10.12.14.101	49254	110.37.219.134	990	252	194 kB	49	106	7 kB	146	186 kB
10.12.14.101	49204	50.23.110.163	80	191	168 kB	1	66	4 kB	125	163 kB
10.12.14.101	49257	187.243.203.67	8090	189	147 kB	52	82	6 kB	107	141 kB
10.12.14.101	49336	110.37.219.134	990	155	119 kB	116	63	6 kB	92	113 kB
10.12.14.101	49295	81.136.148.196	50000	129	108 kB	88	49	4 kB	80	104 kB
10.12.14.101	49330	110.37.219.134	990	135	105 kB	110	58	4 kB	77	100 kB
10.12.14.101	49205	190.146.201.54	80	135	98 kB	2	61	5 kB	74	93 kB
10.12.14.101	49248	217.165.236.108	7080	122	90 kB	43	56	4 kB	66	86 kB
10.12.14.101	49202	103.231.77.68	80	100	62 kB	0	51	4 kB	49	58 kB
10.12.14.101	49249	190.146.201.54	80	85	56 kB	44	44	4 kB	41	52 kB
10.12.14.101	49244	198.61.196.18	8080	61	52 kB	40	21	2 kB	40	50 kB
10.12.14.101	49230	110.37.219.134	990	50	32 kB	22	25	2 kB	25	30 kB
10.12.14.101	49320	110.37.219.134	990	53	28 kB	104	27	3 kB	26	25 kB
10.12.14.101	49282	81.136.148.196	50000	49	28 kB	80	26	3 kB	23	25 kB
10.12.14.101	49301	110.37.219.134	990	40	24 kB	94	19	3 kB	21	22 kB
10.12.14.101	49227	190.146.201.54	80	26	14 kB	19	15	2 kB	11	12 kB
10.12.14.101	49314	110.37.219.134	990	31	14 kB	102	16	2 kB	15	11 kB
10.12.14.101	49233	187.243.203.67	8090	15	8 kB	25	9	1 kB	6	7 kB
10.12.14.101	49225	85.25.211.31	65400	96	7 kB	17	48	4 kB	48	3 kB

Close

Help

Picture 1.4: Sorting by packets to identify the victim (most no of packets is likely the victim)

Wireshark - Follow HTTP Stream (tcp.stream eq 0) - 2018-12-14-Emotet-infection-with-Qakbot.pcap

```

GET /En_us/Clients/122018 HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko
Accept-Encoding: gzip, deflate
Host: tortugadatacorp.com
DNT: 1
Connection: Keep-Alive

HTTP/1.1 301 Moved Permanently
Date: Fri, 14 Dec 2018 16:52:54 GMT
Server: Apache
Location: http://tortugadatacorp.com/En_us/Clients/122018/
Cache-Control: max-age=172800
Expires: Sun, 16 Dec 2018 16:52:54 GMT
Content-Length: 256
Keep-Alive: timeout=5, max=20
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="http://tortugadatacorp.com/En_us/Clients/122018/">here</a>
</p>
</body></html>
GET /En_us/Clients/122018/ HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko
Accept-Encoding: gzip, deflate
Host: tortugadatacorp.com
Connection: Keep-Alive
DNT: 1
  
```

2 client pkt(s), 2 server pkt(s), 3 turn(s).

Entire conversation (96 kB) Show data as ASCII Stream 0

Find: Find Next

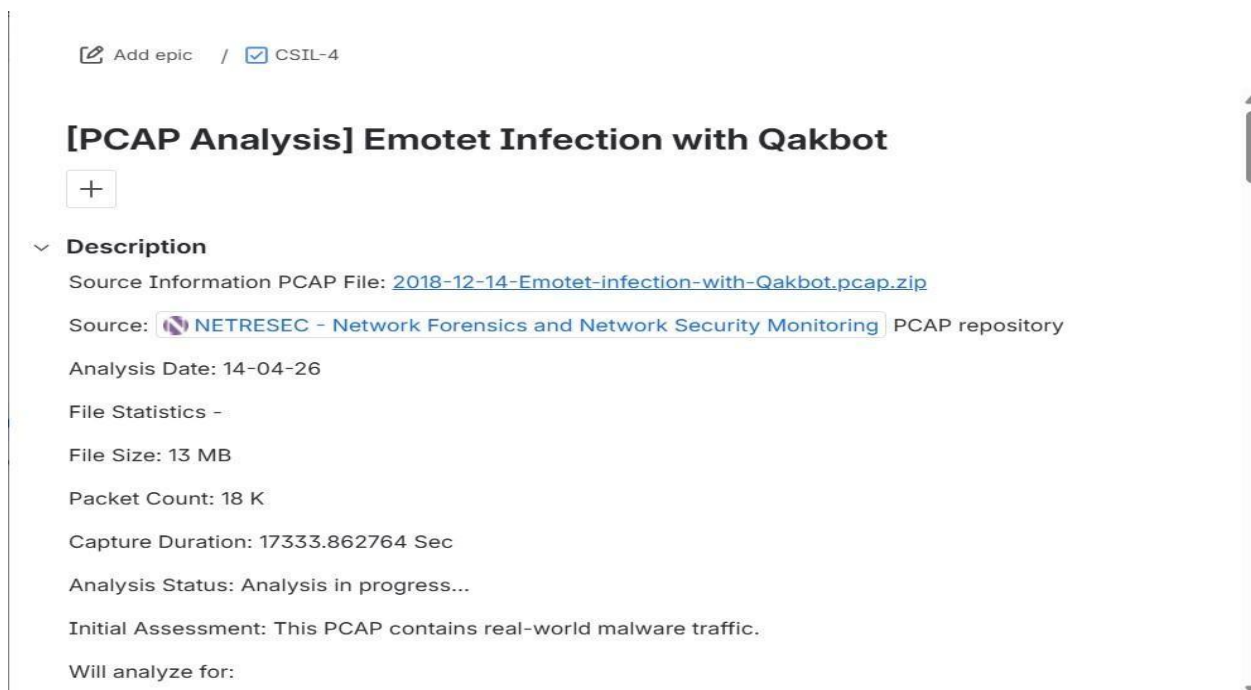
Filter Out This Stream Print Save as... Back Close Help

Picture 1.5: Identified malware download payload by checking http traffic stream of the victim

Jira Ticket — Incident

Detection and Investigation Ticket

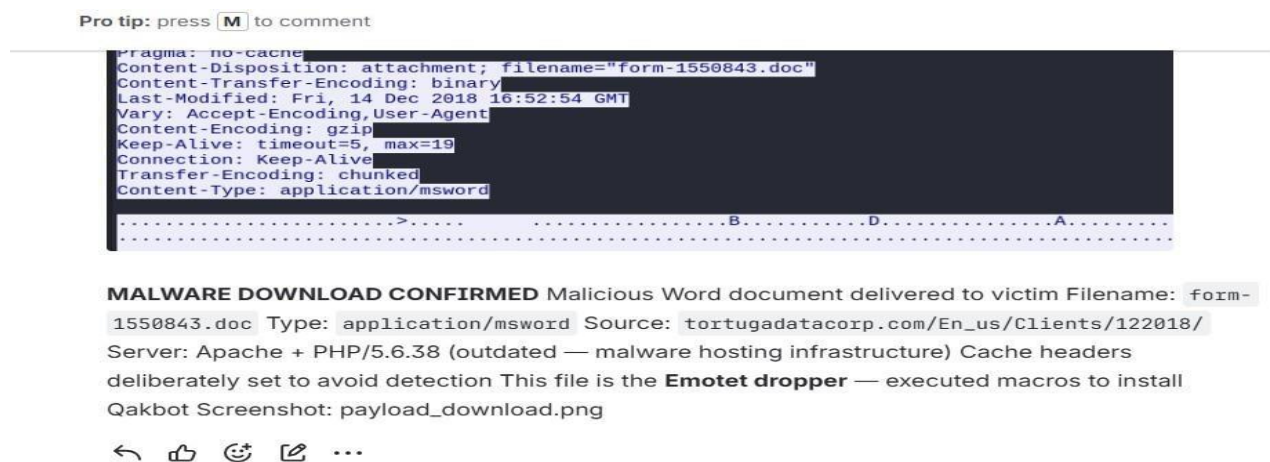
- **Title:** [PCAP Analysis] Emotet Infection with Qakbot
- **Issue Type:** Incident
- **Priority:** High
- **Assignee:** Hadiya Zuberi



The screenshot shows a Jira ticket interface. At the top, there's a link to 'Add epic' and a checkbox for 'CSIL-4'. The ticket title is '[PCAP Analysis] Emotet Infection with Qakbot'. Below the title is a plus icon in a box. The 'Description' section is expanded, showing the following details:

- Source Information PCAP File: [2018-12-14-Emotet-infection-with-Qakbot.pcap.zip](#)
- Source: [NETRESEC - Network Forensics and Network Security Monitoring](#) PCAP repository
- Analysis Date: 14-04-26
- File Statistics -
- File Size: 13 MB
- Packet Count: 18 K
- Capture Duration: 17333.862764 Sec
- Analysis Status: Analysis in progress...
- Initial Assessment: This PCAP contains real-world malware traffic.
- Will analyze for:

Picture 1.5: Detection and investigation Ticket

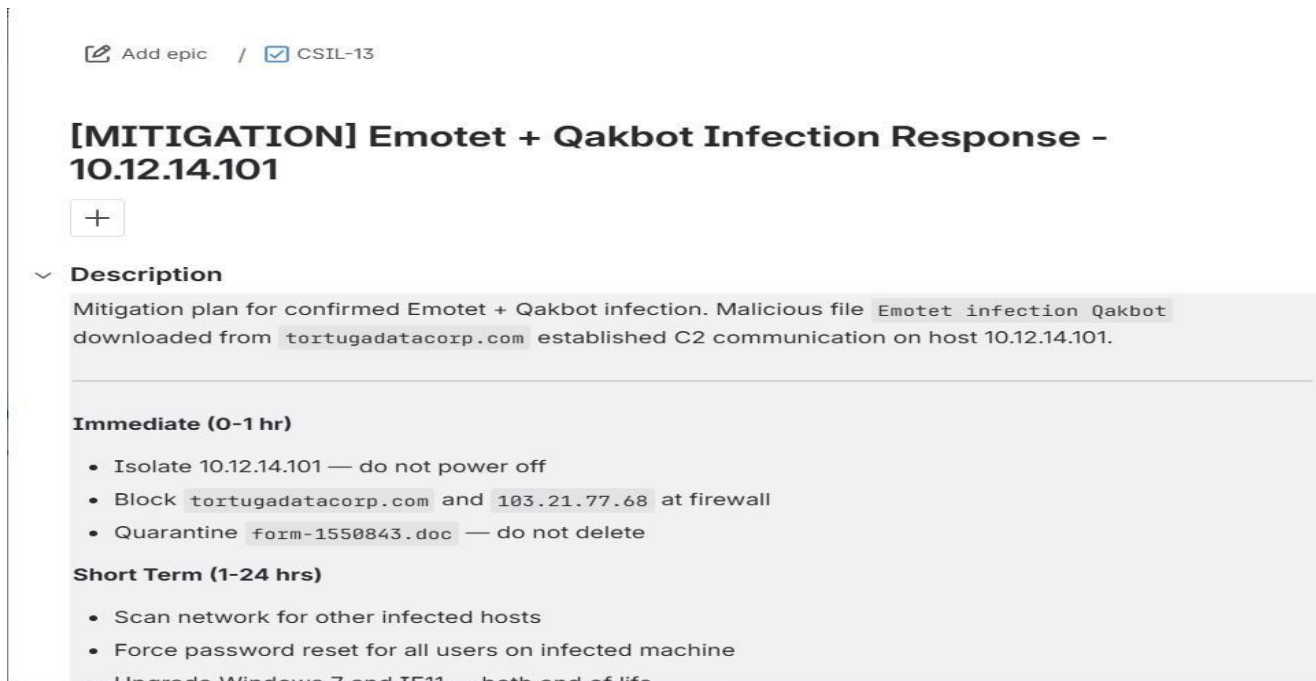


The screenshot shows a Jira comment section. At the top, it says 'Pro tip: press **M** to comment'. Below this is a dark-themed network packet capture analysis. The first part shows HTTP headers for a file named 'form-1550843.doc'. The second part shows a hex dump of the data. Below the hex dump, the text reads: 'MALWARE DOWNLOAD CONFIRMED Malicious Word document delivered to victim Filename: form-1550843.doc Type: application/msword Source: tortugadatacorp.com/En_us/Clients/122018/ Server: Apache + PHP/5.6.38 (outdated — malware hosting infrastructure) Cache headers deliberately set to avoid detection This file is the **Emotet dropper** — executed macros to install Qakbot Screenshot: payload_download.png'. At the bottom, there are icons for replying, liking, reacting, and more options.

Picture 1.6 : Comments for whole analysis process investigation phase

Mitigation Ticket

- **Title:** [MITIGATION] Emotet + Qakbot Infection Response - 10.12.14.101
- **Issue Type:** Incident
- **Priority:** High
- **Assignee:** Hadiya Zuberi
- **Linked to:** [PCAP Analysis] Emotet Infection with Qakbot



Picture 1.7 : Mitigation Ticket linked to detection and investigation

Source & Target IP

Target IP: 10.12.14.101

Source IP: 73.233.236.181

Severity & Status

Priority :High

Status : Closed

Results & Observations

Findings

- DNS Analysis - Suspicious Domains Found

Filter Used: dns

Suspicious Domains Identified:

1. .arpa domain used for internet mapping, primarily for reverse DNS lookups (mapping IP addresses back to domain names).

- TLS/SSL Analysis - C2 Discovery

Filter Used: tls.handshake.type == 1

1. Infected machine repeatedly connects to 72.233.236.161
- HTTP Redirect to Malicious Domain Confirmed
1. External IP 103.21.77.68 redirected victim (10.12.14.101) to tortugadatacorp.com/En_us/Clients/122018/

MALWARE DOWNLOAD CONFIRMED

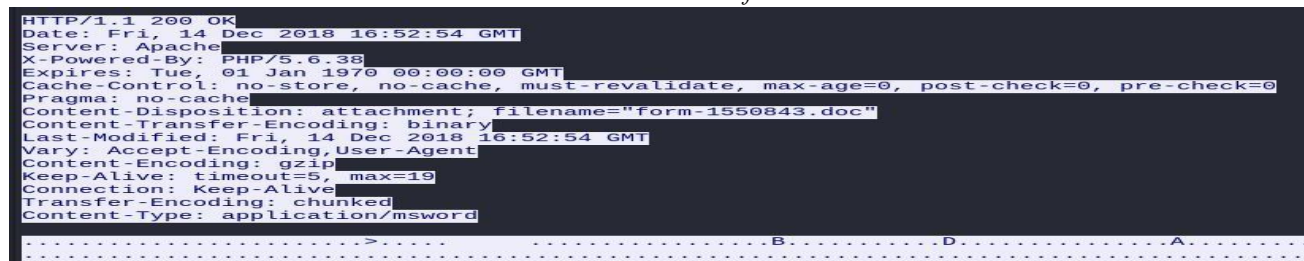
Malicious Word document delivered to victim Filename: form-1550843.doc Type: application/msword Source: tortugadatacorp.com/En_us/Clients/122018/ Server: Apache + PHP/5.6.38 (outdated — malware hosting infrastructure) Cache headers deliberately set to avoid detection This file is the **Emotet dropper** — executed macros to install Qakbot Screenshot: payload_download.png

Screenshots

IOCs to Block

Type	Value
Domain	tortugadatacorp.com
IP	103.21.77.68
File	form-1550843.doc

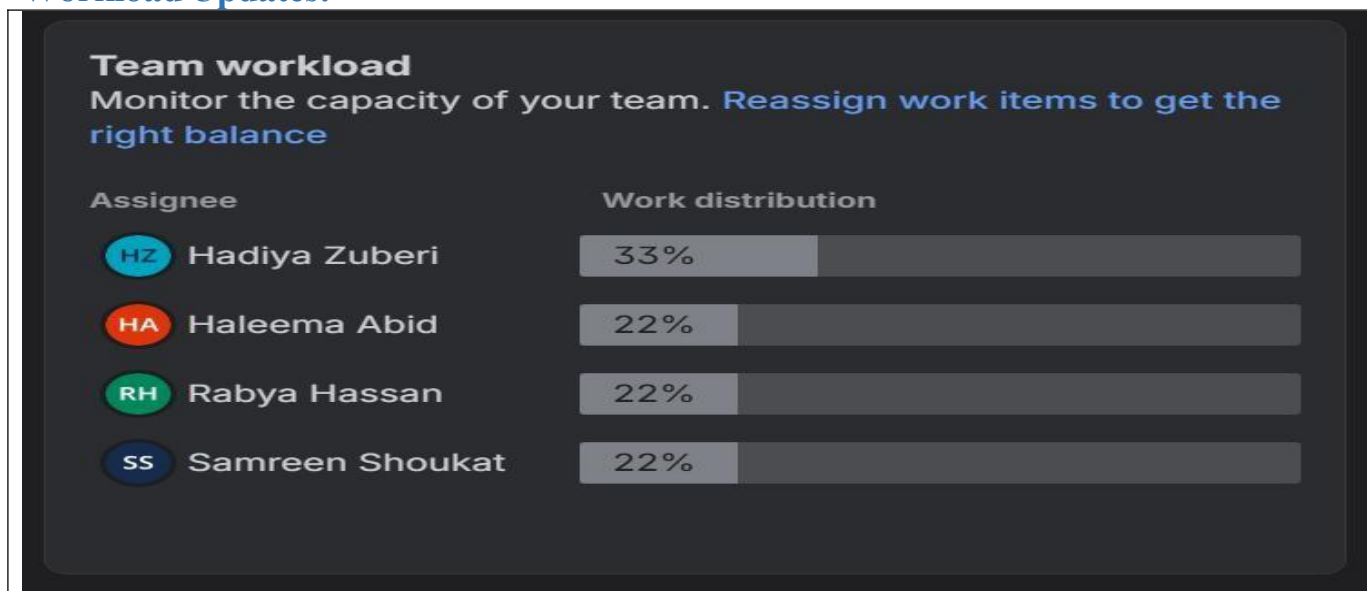
Picture 1.8: IOC's identified



Picture 1.9: Malicious Payload

WorkFlows

Workload Updates:



List Updates:

☐	☒ CSIL-13 [MITIGATION] Emotet + Qakbot Infection Re...	 Unassigned	 Hadiya Zuberi	= Medium
☐	☒ CSIL-12 Mitigation	 Rabya Hassan	 Samreen Shoukat	= Medium
☐	☒ CSIL-11 Investigation	 Unassigned	 Samreen Shoukat	= Medium
☐	☒ CSIL-10 Investigation — MitM Injection Mechanism	 Hadiya Zuberi	 Samreen Shoukat	= Medium
☐	☒ CSIL-9 Malware Redirect URLs Identified	 Haleema Abid	 Samreen Shoukat	= Medium
☐	☒ CSIL-8 MitM HTTP Injection Detected	 Samreen Shoukat	 Samreen Shoukat	= Medium
☐	☒ CSIL-7 Mitigation:SYN Flood Attack	 Hadiya Zuberi	 Haleema Abid	= Medium
☐	☒ CSIL-6 Investigation:SYN Flood Attack	 Haleema Abid	 Haleema Abid	^ High
☐	☒ CSIL-5 Syn Flood Attack Detected	 Samreen Shoukat	 Haleema Abid	^ High
☐	☒ CSIL-4 [PCAP Analysis] Emotet Infection with Qakbot	 Hadiya Zuberi	 Hadiya Zuberi	= Medium
☐	☒ CSIL-3 Port Scanning Detector	 Rabya Hassan	 Haleema Abid	= Medium
☐	☒ CSIL-1 Analysis	 Unassigned	 Haleema Abid	= Medium

Status Updates:

Spaces

Cyber Security Incident Lab

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

Archived work items

Status overview

Get a snapshot of the status of your work items. [View all work items](#)

12

Total work item...

resolved: 9

Open: 3

Recent activity

Stay up to date with what's happening across the space.

Today

Hadiya Zuberi updated field "status" on ☒ CSIL-4: [PCAP Analysis] E

motet Infection with Qakbot

RESOLVED

7 minutes ago

Hadiya Zuberi updated field "status" on ☒ CSIL-4: [PCAP Analysis] E

motet Infection with Qakbot

RESOLVED

8 minutes ago

Samreen Shoukat updated field "Sprint" on ☒ CSIL-13: [MITIGATION]

Emotet + Qakbot Infection Response - 10.12.14.101

OPEN

Priority breakdown

Get a holistic view of how work is being prioritized. [How to manage priorities for](#)

Types of work

Get a breakdown of work items by their types. [View all items](#)

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